

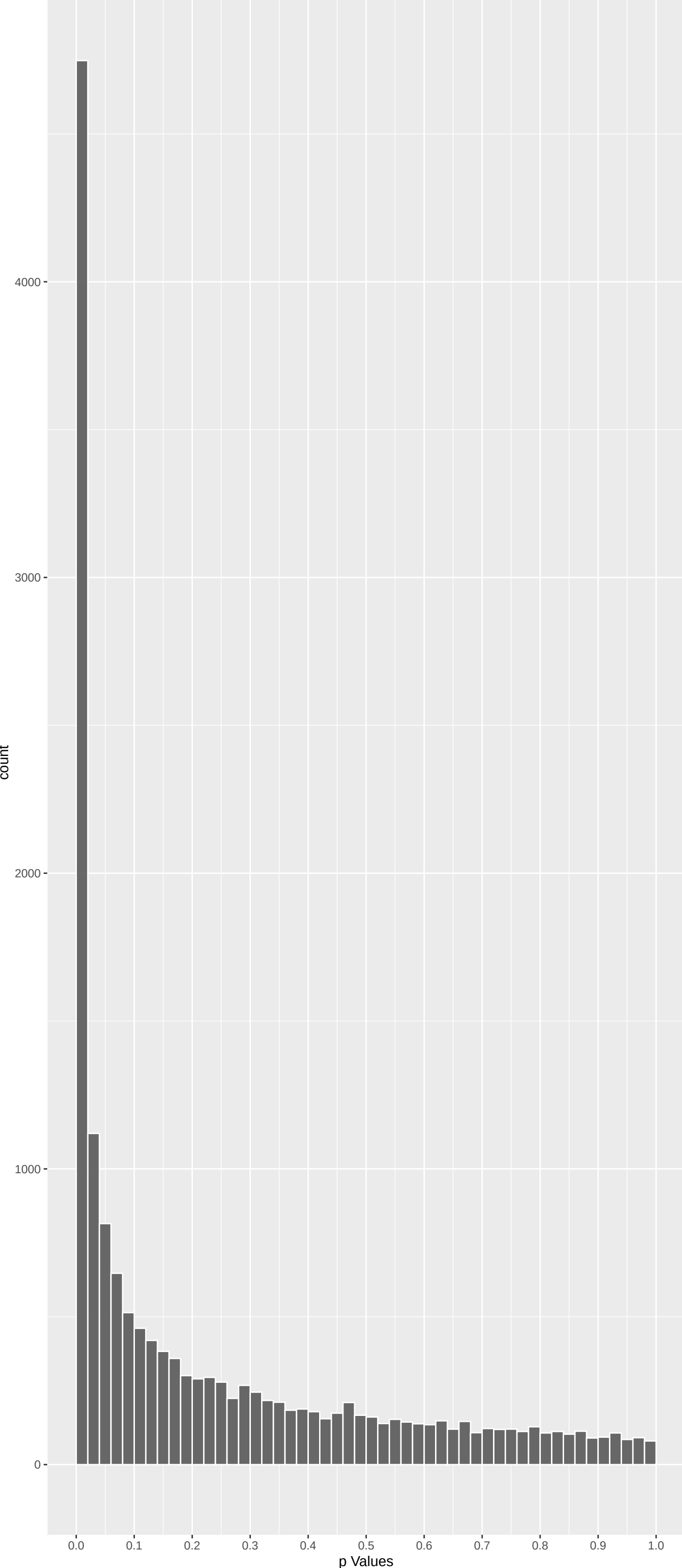
Top genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
TRIM15	0.17618490	441	1.044e-16	1.673e-12	7.130e-09
DNAH14	0.13788425	548	2.941e-16	2.357e-12	7.130e-09
EP400	0.11287714	602	1.238e-14	6.616e-11	1.334e-07
SPG11	0.09823555	669	3.694e-14	1.377e-10	1.463e-07
TRPM5	0.12162818	538	4.297e-14	1.377e-10	1.463e-07
ADGRF1	0.10115102	642	5.433e-14	1.451e-10	1.463e-07
TMEM131L	0.10042259	625	1.585e-13	3.629e-10	3.136e-07
PLEC	0.12854499	477	3.277e-13	6.566e-10	4.965e-07
ZFAND4	0.10795991	549	8.877e-13	1.581e-09	1.063e-06
MTBP	0.09459793	620	1.196e-12	1.917e-09	1.160e-06
PGBD1	0.12357687	458	3.327e-12	4.848e-09	2.380e-06
HIRA	0.13640378	408	5.328e-12	5.902e-09	2.380e-06
PJA2	0.10619452	526	4.745e-12	5.902e-09	2.380e-06
KIAA0753	0.09302514	597	5.523e-12	5.902e-09	2.380e-06
SFSWAP	0.10479511	531	5.263e-12	5.902e-09	2.380e-06
HERC2	0.09522954	564	1.356e-11	1.047e-08	2.879e-06
AKAP9	0.08600872	629	1.115e-11	1.047e-08	2.879e-06
ECI2	0.09634744	559	1.261e-11	1.047e-08	2.879e-06
TGFBRAP1	0.09911193	541	1.420e-11	1.047e-08	2.879e-06
ADGRG4	0.13340688	402	1.437e-11	1.047e-08	2.879e-06
FGD5	0.09892477	546	1.170e-11	1.047e-08	2.879e-06
RIMBP2	0.09392215	573	1.285e-11	1.047e-08	2.879e-06
BIRC3	0.10530464	505	1.768e-11	1.232e-08	3.209e-06
PPP4R1	0.09928807	534	1.906e-11	1.273e-08	3.209e-06
DYNC1LI1	0.13401090	394	2.160e-11	1.385e-08	3.351e-06

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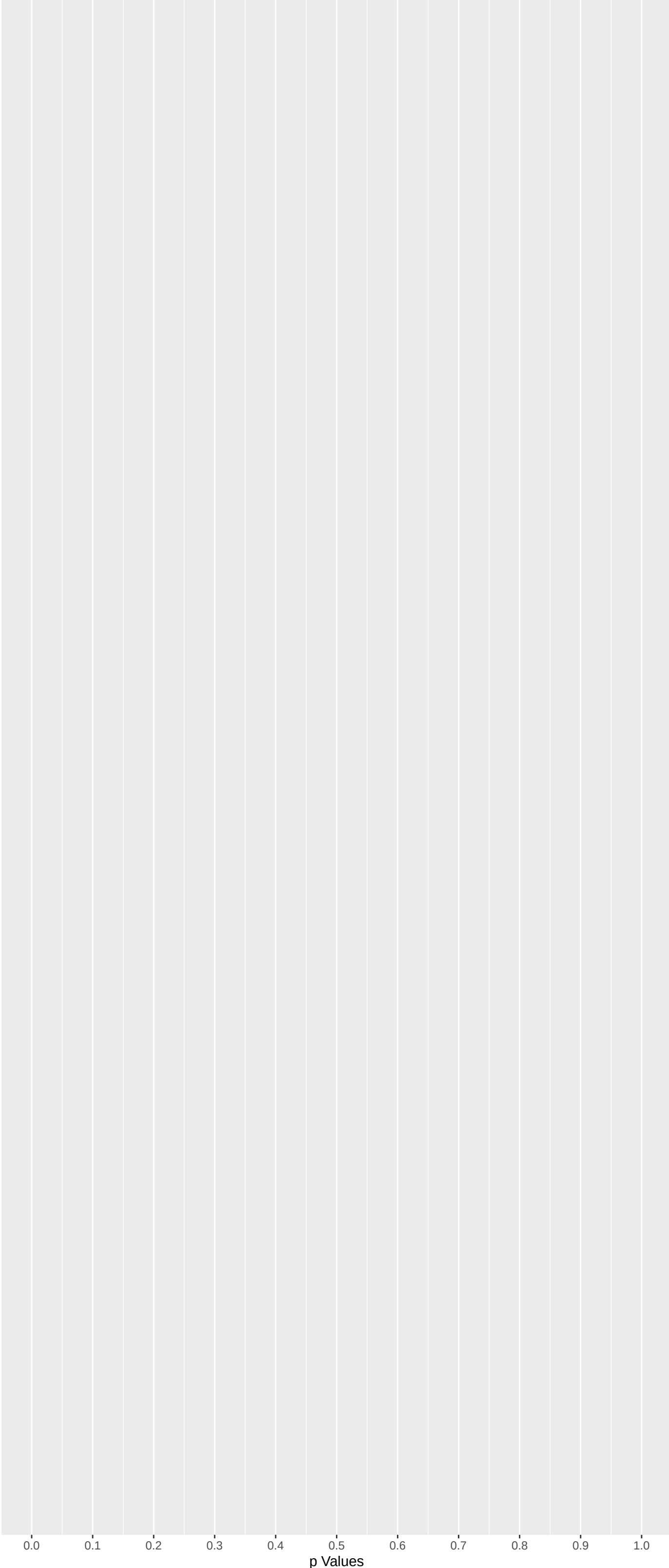
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

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TRIM15	0.17618490	441	1.044e-16	1.673e-12	7.130e-09
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ZFAND4	0.10795991	549	8.877e-13	1.581e-09	1.063e-06
MTBP	0.09459793	620	1.196e-12	1.917e-09	1.160e-06

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
mRNA Processing (GO:0006397)	-0.11395316	178	1.707e-07	9.196e-04	SF3B3;6 BUD31;44 IK;120 NCBP2;122 RBM10;178 RNF113A;317
RNA Splicing, Via Transesterification Re (GO:0008380)	-0.12051181	149	4.081e-07	1.099e-03	SF3B3;6 BUD31;44 IK;120 NCBP2;122 RBM10;178 RNF113A;317
RNA Splicing (GO:0008380)	-0.15573798	81	1.302e-06	1.800e-03	SF3B3;6 NCBP2;122 RNF113A;317 GEMIN2;351 HNRNP;3;466 PSC;653
mRNA Splicing, Via Spliceosome (GO:00003)	-0.10477897	180	1.336e-06	1.800e-03	SF3B3;6 BUD31;44 GEMIN4;59 IK;120 NCBP2;122 RBM10;178
Positive Regulation Of Transcription By (GO:0027350)	-0.04583600	873	6.053e-06	6.523e-03	TAFO10;22 E2F3;31 HAND1;83 HNF4A;84 USF2;87 PSMC3;91
Gene Expression (GO:0010467)	-0.08227350	250	8.163e-06	7.331e-03	DNAJB11;19 NCBP2;122 ABCF1;138 NLRP3;155 HNRNP;171 MRPL55;176
RNA Processing (GO:0006396)	-0.10166408	158	1.090e-05	8.89e-03	SF3B3;6 NCBP2;122 HNRNP;171 DEDD2;315 GEMIN2;351 EXOSC6;421
Positive Regulation Of DNA-templated Tra (GO:0037827)	-0.03758267	1149	2.608e-05	1.756e-02	SF3B3;6 TAFO10;22 E2F3;31 ACTR18;36 FLN1;40 MSL;368
Inorganic Cation Transmembrane Transport (GO:007121956)	0.07121956	277	4.903e-05	2.935e-02	TRPM5;5 ATP6V0A2;43 CNKNI5;80 CNKN18;125 TRPM6;143 ATP2A1;153
Macromolecule Biosynthetic Process (GO:000932725)	-0.09392725	147	8.799e-05	4.557e-02	LARGE1;66 ABCF1;138 MRPL55;176 MRPL3;273 MRPS18B;284 MRPL27;338
Vesicle Budding From Membrane (GO:000690)	-0.20623063	30	9.303e-05	4.57e-02	VPS4A;90 SNX3;294 S100A10;359 GOLPH3;433 TRAPP5;500 FNPB1;931
Monatomic Cation Transmembrane Transpor (GO:006846056)	0.06846056	273	1.065e-04	4.782e-02	TRPM5;5 ATP6V0A2;43 CNKNI5;177 RCLH18;125 TRPM6;143 ATP2A1;153
Monocarboxylic Acid Transport (GO:001571)	0.13248030	63	2.798e-04	9.424e-02	ATP8B1;113 MFS02A;129 ABCC11;177 RCLH1;1223 ZFPK6;197 CNOT7;1933
piRNA Processing (GO:0034587)	0.24814489	18	2.682e-04	9.424e-02	MOV10L1;186 SPAT2;406 PIWILL;607 TEX15;894 ANKRD34C;954 PNLD;1;989
Protein Localization To Centrosome (GO:000506837)	0.25506837	17	2.719e-04	9.424e-02	KIAA0753;15 CEP192;107 STIL;522 GOLGB1;948 SPAG5;950 NUDCD3;1173
Translation (GO:0006412)	-0.07651533	194	2.494e-04	9.424e-02	MTG1;126 ABCF1;138 MRPL55;176 APEH;224 MRPL3;273 MRPS18B;284
B Cell Homeostasis (GO:0001782)	-0.33637994	9	5.400e-04	1.608e-01	PPP2R3C;267 GAPT;289 TNFRSF13B;693 TNFAIP3;1505 MEF2C;2248 DOCK10;2506
Peptide Biosynthetic Process (GO:0043043)	-0.09089664	122	5.386e-04	1.608e-01	MRPL55;176 GGT1;233 MRPL3;273 MRPS18B;284 MRPL27;338 RPLP0;406
Regulation Of Small GTPase Mediated Sign (GO:00529467)	0.00529467	110	5.672e-04	1.608e-01	FGD5;17 TIAM2;180 SYDE2;197 ARHGEF19;224 FAMI13B;410 MCF2L;640
Positive Regulation Of Interferon-Beta P (GO:00787322)	-0.17539232	32	5.982e-04	1.612e-01	RIOK3;333 IRF5;779 TLR4;803 TRIM56;988 HSP90AA1;1107 IRF3;1194
Intermediate Filament Organization (GO:00013939306)	0.13939306	50	6.557e-04	1.627e-01	KRT80;408 KRT32;470 KRT73;572 KRT5;1007 KRT19;1009 BFSF2;1226
Regulation Of Transcription By RNA Polym (GO:002565134)	-0.02565134	1698	6.645e-04	1.627e-01	SF3B3;6 TAFO10;22 E2F3;31 HOXA13;32 FLN1;40 ERF;43
Organic Hydroxy Compound Transport (GO:00016087499)	0.16087499	36	8.415e-04	1.971e-01	ATP8B1;113 ABCC11;177 SLC10A2;1147 GHRL;1248 SLC10A6;1249 ABCC3;1403
Regulation Of DNA-templated Transcription (GO:00516206)	-0.02516206	1636	1.014e-03	2.275e-01	SF3B3;6 TAFO10;22 E2F3;31 HOXA13;32 ACTR18;36 FLN1;40
Cellular Respiration (GO:0045333)	-0.11187876	71	1.152e-03	2.484e-01	COX4I1;12 CHCHD5;61 NDUVF1;253 UQCNR3;1;680 ME3;799 NDUFS8;836
Non-Motile Cilium Assembly (GO:1905515)	0.16700886	31	1.295e-03	2.674e-01	ARL13B;213 IFT122;464 CEP89;1302 TOGARAM1;1396 FUZ;1501 CC2CD3;1691
Protein Localization To Endoplasmic Retic (GO:00370667)	-0.22657935	6	1.355e-03	2.674e-01	GBF1;421 SEC16B;734 SEC16A;1389 MIA3;2344 LRRK1;343 CBAP31;3496
Response To Lipopolysaccharide (GO:00324)	-0.08263328	126	1.390e-03	2.674e-01	CD6;9 NLRP3;155 CD274;300 SHPK;309 RELA;370 SBNQ2;372
Protein Peptidyl-Prolyl Isomerization (GO:00454523)	-0.24545423	14	1.618e-03	2.741e-01	PP1F;158 PP1L3;397 PP1G;653 PP1D;773 PP1L6;1419 PP1A;1851
Metal Ion Transport (GO:0030001)	0.07309780	157	1.618e-03	2.812e-01	CLDN16;52 SLC17A4;75 CNKNI5;179 RCLH13;95 KCNK18;125 ATP2A1;153
Regulation Of Nuclear-Transcribed mRNA C (GO:00347075)	-0.19442705	22	1.599e-03	2.812e-01	NANOS3;179 SCDE1;265 NANCOS1;72 RCLH1;1223 ZFPK6;197 CNOT7;1933
Epithy Involved In Wound Healing (GO:0002448381)	-0.24748381	13	2.006e-03	2.844e-01	TMEEFF2;687 DDR1;1467 CD151;1596 FLNA;1717 CD44;2339 LCP1;2626
Placenta Development (GO:0001890)	0.22346878	16	1.972e-03	2.844e-01	PTK2;906 SOD1;1091 PPARG;1275 LEP;2039 ESRRB;2587 PKD2;3366
Positive Regulation Of Protein Binding (GO:00011992828)	-0.11992828	56	1.923e-03	2.844e-01	NMD3;2 RAN;11 BLK;134 TRIM21;225 DERL1;451 NVL;716
Positive Regulation Of Type I Interferon (GO:00012155942)	-0.12155942	55	1.834e-03	2.844e-01	RIOK3;333 RAB2B;714 IRF5;779 TLR4;803 TRIM56;988 TRAF3;1082
Potassium Ion Transmembrane Transport (GO:000787322)	0.0787322	132	1.794e-03	2.844e-01	KCNJ16;79 CNKNI5;80 CNKN18;125 KCNV2;334 CNKNI29;9 KNKNA;364
Protein Localization To Microtubule Orga (GO:023109706)	0.23109706	15	1.945e-03	2.844e-01	KIAA0753;15 CEP192;107 STIL;522 SPAG5;950 CC2CD3;1691 CEP131;2475
rRNA Metabolic Process (GO:0016072)	-0.10480197	75	1.722e-03	2.844e-01	SBD5;236 SPLP2B;298 DEDD2;315 EXOSC6;421 ESF1;670 XRN2;690
Amino Acid Transport (GO:0006865)	0.13486711	43	2.227e-03	2.950e-01	SLC43A1;105 SLC6A6;220 SLC6A20;713 SLC6A5;835 SLC36A2;1017 SLC7A4;1229
Cellular Response To Molecule Of Bacteri (GO:009361040)	-0.09361040	90	2.176e-03	2.950e-01	NLRP3;155 CD274;300 SHPK;309 RELA;370 SBNQ2;372 NFKB1;388

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HOUNKPE_HOUSEKEEPING_GENES	-0.08685315	1032	9.561e-21	6.203e-17	NMD3;2 STK38;7 MAPRE1;10 SARI1;17 DNAJB11;19 TAFO10;22
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.07327350	1126	2.308e-16	7.486e-13	COX4I1;12 GNK14;24 RSU1;25 TUBG2;39 LARGE1;66 CKMT1B;69
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	-0.05675721	1292	1.298e-11	2.808e-08	NMD3;2 SF3B3;6 CEP70;13 RSU1;25 FSCN1;26 ACOT9;35
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	-0.22657935	68	1.060e-10	1.720e-07	MAPRE1;10 PSMC3;91 CCNB1;157 PSMB1;210 TUBB4A;344 PSMF1;529
REACTOME_METABOLISM_OF_RNA	-0.07582590	608	2.160e-10	2.803e-07	SF3B3;6 RAN;11 NHP2;23 BUD31;44 GEMIN4;59 PSMC3;91
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	-0.05774115	987	1.184e-09	1.098e-06	COX4I1;12 ZNF624;14 TAFO10;22 PAF1;60 NPY;76 CDC27;82
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	-0.06964808	667	1.083e-09	1.098e-06	COX4I1;12 DNAJB11;19 E2F3;31 ERF;43 CDC27;82 PSMC3;91
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	-0.18735968	78	1.080e-08	8.761e-06	PSMC3;91 ANP32A;139 PSMB1;210 TNFSF13;319 SET;332 EXOSC6;421
REACTOME_INFECTIOUS_DISEASE	-0.06038195	781	1.215e-08	8.761e-06	RAN;11 TAFO10;22 GEMIN4;59 ADCY6;86 VPS4A;90 PSMC3;91
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.12375431	177	1.420e-08	9.215e-06	PSMC3;91 BLK;134 PSMB1;210 CSNK2B;215 HNCB1;364 THBS1;440
MILI_PSEUDOPODIA_HAPTOTAXIS_UP	-0.07523902	469	2.713e-08	1.600e-05	RGS20;118 PLEKH48;149 AKK1;653 DTNA;160 CENP1;161 DUCND1D1;238
KIM_BIPOLAR_DISORDER_OLIGODENDROCYTE_DEN	-0.06591908	609	3.309e-08	1.789e-05	RAN;11 TAFO10;22 NPY;76 TSNAK;103 CA11;109 SPTBN2;113
REACTOME_REGULATION_OF_EXPRESSION_OF_SLI	-0.14123350	127	3.998e-08	1.995e-05	PSMC3;91 NCBP2;122 PSMB1;210 RPLP0;406 GSP1T;1437 PSMF1;529
REACTOME_MRNA_SPLICING	-0.11773976	182	4.498e-08	2.085e-05	SF3B3;6 BUD31;44 IK;120 NCBP2;122 HNRNP;171 RBM10;178
BENPORATH_MYC_MAX_TARGETS	-0.06099854	675	7.875e-08	3.406e-05	MAPRE1;10 POLR1B;18 NCBP2;122 NUDT6;132 BLK;134 DNAI1;147
SPIELMAN_LYMPHOBLAST_EUROPEAN_VS_ASIAN_D	-0.06641996	541	1.455e-07	5.901e-05	NMD3;2 MAPRE1;10 PSMK1;52 GEMIN4;59 CDC27;82 EXOSC6;111
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	-0.11633168	169	1.886e-07	6.439e-05	PSMC3;91 NCBP2;122 CMBP1;210 RPLP0;406 GSP1T;1437 PSMF1;529
REACTOME_UCH_PROTEINASES	-0.16885480	80	1.807e-07	6.439e-05	ACTR18;36 PSMC3;91 USP15;117 PSMB1;210 TFFT;228 OGT;343
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	-0.16884998	80	1.808e-07	6.439e-05	PSMC3;91 PSMB1;210 HAC1;354 PSMF1;529 PSMA;7;660 HGCL1;1041
REACTOME_INNATE_IMMUNE_SYSTEM	-0.05269506	849	2.312e-07	7.501e-05	RAB3D;34 PSMC3;91 POLR3;1105 C8A;140 NFKBIA;146 NLRP3;155
PUJANA_BRCA1_PCC_NETWORK	-0.04111113	1448	2.441e-07	7.542e-05	ITPK1;8 MAPRE1;10 RAN;11 NHP2;23 E2F3;31 BUD31;44
MARSON_BOUND_BY_FOXP3_UNSTIMULATED	-0.06411546	1098	3.323e-07	8.936e-05	SPA17;1 LYRM4;4 DET1;15 UTP23;28 FLN1;40 FCRLA;48
ACEVEDO_LIVER_TUMOR_VS_NORMAL_ADJACENT_T	-0.05454248	763	3.581e-07	8.936e-05	MAPRE1;10 DNAJB11;19 NHP2;23 ACOT9;35 GID8;77 LSM10;101
MARTENS_TRETINOIN_RESPONSE_DN	-0.05772435	686	3.038e-07	8.936e-05	RAN;11 COX4I1;12 GEMIN4;59 NTSY5;729 ESD;164 HNRNP;171
KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER	-0.05754311	682	3.566e-07	8.936e-05	RAN;11 COX4I1;12 TAFO10;22 NPY;76 TSNAK;103 CA11;109
HSIAO_HOUSEKEEPING_GENES	-0.08388068	316	3.176e-07	8.936e-05	ITPK1;8 COX4I1;12 BUD31;44 NFKBIA;146 ESD;164 HNRNP;171
REACTOME_TCR_SIGNALING	-0.14730121	98	4.769e-07	1.146e-04	PSMC3;91 NFKBIA;146 PSMB1;210 RELA;370 NFKB1;388 CD101;515
REACTOME_CLEC7A_DECTIN_1_SIGNALING	-0.15675366	86	5.123e-07	1.187e-04	PSMC3;91 NFKBIA;146 PSMB1;210 PPP3CA;251 RELA;370 NFKB1;388
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	-0.14737998	97	5.374e-07	1.202e-04	PSMC3;91 BLK;134 NFKBIA;146 CDRF;9;163 PSMB1;210 PPP3CA;251
DANG_BOUND_BY_MYC	-0.04896730	920	5.996e-07	1.261e-04	MAPRE1;10 POLR1B;18 E2F3;31 ERF;43 NCBP2;122 NUDT6;132
CHIANG_LIVER_CANCER_SUBCLASS_UNANNOTATED	-0.10860164	178	6.024e-07	1.261e-04	MAPRE1;10 RAN;11 PSMC3;91 UBE2E3;119 HAUS1;166 CIAO2A;195
KEGG_SPLICEOSOME	-0.14374847	100	6.924e-07	1.404e-04	SF3B3;6 BUD31;44 NCBP2;122 CHERP;598 SNU13;707 SRSF6;940
DACOSTA_UV_RESPONSE_VIA_ERCC3_UP	-0.08675967	276	7.474e-07	1.469e-04	IGFBP6;50 PSMC3;91 NCBP2;122 PP1F;158 TRIM21;225 NDUVF1;253
PUJANA_CHEK2_PCC_NETWORK	-0.05557357	686	8.212e-07	1.567e-04	RAN;11 SARI1;17 NHP2;23 NCBP2;122 GTF2E1;141 RAD51C;156
BRIUNIS_UVC_RESPONSE_LATE	-0.04538889	1040	9.633e-07	1.777e-04	SPA17;1 MMP23B;3 RSU1;25 ACOT9;35 CKMT1B;69 NTS5C2;79
LOPEZ_MBD_TARGETS	-0.05044816	828	9.900e-07	1.777e-04	RAN;11 FLN1;40 PSMC3;91 TNC;96 CCS;114 NFKBIA;146
BORCZUK_MALIGNANT_MESOTHELIOMA_UP	-0.08511041	280	1.014e-06	1.777e-04	MAPRE1;10 RAN;11 NHP2;23 FSCN1;26 CDC27;82 UBE2E3;119
KIM_ALL_DISORDERS_CALB1_CORR_UP	-0.06328999	510	1.101e-06	1.880e-04	RAN;11 NPY;76 TSNAK;103 UBE2E3;119 PDHX;148 DDOST;1
REACTOME_POST_TRANSLATIONAL_PROTEIN_MODI	-0.04123162	1253	1.241e-06	2.064e-04	PGM3;16 TAFO10;22 RAB3D;34 ACTR18;36 COMM5;37 PAF1;60
REACTOME_PROTEIN_FOLDING	-0.15829590	78	1.361e-06	2.207e-04	GNAI4;24 CSNK2B;215 TUBB4A;344 GNAQ;528 TUBB4B;651 XRN2;690

DisGeNET Top pathways by non-permutation

Cytochrome-c Oxidase Deficiency	-0.21433922	36	8.647e-06	8.483e-02	COX4I1;12 CCS;114 COA7;218 COA5;242 PINK1;350 FARS2;550
Glioma	-0.03022869	1763	4.457e-05	2.186e-01	RSU1;25 FSCN1;26 E2F3;31 HOXA13;32 IGFBP6;50 CMKP1;52
Epileptic encephalopathy	-0.05489292	439	9.038e-05	2.217e-01	LARGE1;66 SLC1A2;70 TPP1;98 PDHX;148 TFE3;169 AKT3;181
Tetany	0.25046515	21	7.110e-05	2.217e-01	HIRA;14 CLDN16;52 TRPM6;143 JMJD1C;618 KCNJ1;800 CASR;851
Anemia, hereditary spherocytic hemolytic	0.47822789	5	2.127e-04	2.230e-01	SPTA1;74 ANK1;76 SLC4A1;287 SPTB;483 EPB42;807 NA
Central neuroblastoma	-0.03037597	1394	2.086e-04	2.230e-01	RAN;11 SLC25A27;29 E2F3;31 FLN1;40 IGFBP6;50 MSX1;67
Glioblastoma	-0.02859081	1581	2.273e-04	2.230e-01	RSU1;25 E2F3;31 HOXA13;32 MSX1;67 NPY;76 TNC;96
Metabolic Bone Disorder	0.19862091	30	1.672e-04	2.230e-01	CYP2R1;217 CA2;590 MMP2;1139 KDM6B;1209 ANKRD11;1361 PTH;1567
Spherocytosis	0.47822789	5	2.127e-04	2.230e-01	SPTA1;74 ANK1;76 SLC4A1;287 SPTB;483 EPB42;807 NA
Squamous cell carcinoma	-0.02942937	1485	2.213e-04	2.230e-01	SPA17;1 MAPRE1;10 UPP2;21 FSCN1;26 E2F3;31 HPSE2;42
Lactic acidemia	-0.09568405	120	3.018e-04	2.692e-01	LYRM4;4 PDHX;148 COA7;218 COA5;242 NDUVF1;253 SUCLA1;455
22q11 Deletion Syndrome	0.23662784	19	3.567e-04	2.692e-01	HIRA;14 CHRND;428 JMJD1C;618 FGF8;1111 ZDHHC8;1326 ARVCF;1938
Precursor T-cell Lymphoblastic Leukemia-	-0.05746331	333	3.374e-04	2.692e-01	CD6;9 HOXA13;32 MSX1;67 LBL;181 XP150;229 NTSN;191
Familial intrahepatic cholestasis of pre	0.14802232	6	3.910e-04	2.740e-01	ATP8B1;113 ABCB4;369 VCAM1;581 NR1H4;1935 ABCB11;1965 TNF;2790
Polyarticular Juvenile Idiopathic Arthri	-0.23556095	18	5.413e-04	3.123e-01	IL2RB;818 CCR3;1075 STAT4;1166 JAZF1;1371 IL2RA;1374 IL6R;1381
Classical Hodgkin's Lymphoma	-0.08498580	142	4.857e-04	3.123e-01	FSCN1;26 NFKBIA;146 CYBB;167 CD274;300 NFKB1;388 FOXO1;446
Skin Manifestations	0.30154491	11	5.345e-04	3.123e-01	PLEC;8 ACAD8;174 CYP11A1;820 NTS1;1074 VEGFA;2035 IL33;2246
Neuroblastoma	-0.02787960	1432	5.734e-04	3.125e-01	LYRM4;4 RAN;11 SLC25A27;29 E2F3;31 FLN1;40 IGFBP6;50
Acidosis, Lactic	-0.08034857	150	7.022e-04	3.131e-01	LYRM4;4 PDHX;148 COA7;218 COA5;242 NDUVF1;253 SUCLA1;455
Neoplasm Metastasis	-0.02028659	3114	6.476e-04	3.131e-01	MMP23B;3 RAN;11 CEP70;13 UPP2;21 RSU1;25 FSCN1;26
Nephrocalcinosis	0.13492728	54	6.096e-04	3.131e-01	CLCNKB;48 CLDN16;52 TRPM6;143 NSD1;169 CYP2R1;217 MGRFRP;229
Nuclear cataract	0.17364322	32	6.780e-04	3.131e-01	ERC6C;412 GJA8;639 HSFA;741 TGM2;860 GJA3;885 CRYBA4;1262
Ostium secundum atrial septal defect	-0.20597992	22	8.265e-04	3.440e-01	MSX1;67 CITED2;793 TAB2;1382 F2;1566 HTPN;11591 PQBP1;1693
Proluria	0.39223891	6	8.768e-04	3.440e-01	SLC6A20;713 SLC36A2;1017 ALDH4A1;1668 OPLAH;1860 SLC6A19;2326 PRODH;2581
Thick vermilion border	-0.13923546	48	8.522e-04	3.440e-01	ERF;43 DSPRY1;292 NAGA;418 AMMECR1;586 CDK16;619 SNX14;735
Ciliopathies	0.07277440	170	1.100e-03	3.720e-01	KIA00753;15 RSNC;138 ARL13B;213 FAM161A;232 LZTFL1;297 INVS;438
Denys-Drash Syndrome	0.31567533	9	1.041e-03	3.720e-01	APP;624 MIT1;1029 EGF;1052 VEGFA;2035 EGRF;259 PKD3;3366
Hamartoma of tongue	0.42254634	5	1.067e-03	3.720e-01	OFD;1854 NEK1;1042 WDPCP;1121 TCN3;1317 C2CD3;1691 NA
Malignant neoplasm of stomach	-0.02371777	1859	1.057e-03	3.720e-01	STK38;7 MAPRE1;10 GN1A;24 FSCN1;26 E2F3;31 HOXA13;32
Bilateral cataracts (disorder)	0.09141891	103	1.370e-03	4.193e-01	GSTO1;88 GJA8;639 TDRD7;653 SMN2;623 FHF4;71 CRYGA;838
Liver carcinoma	-0.01968117	2781	1.470e-03	4.193e-01	MMP23B;3 MAPRE1;10 RAN;11 GN1A;24 FSCN1;26 E2F3;31
Non-Small Cell Lung Carcinoma	-0.02371226	1728	1.496e-03	4.193e-01	SPA17;1 MAPRE1;10 FSCN1;26 IGFBP6;50 CMKP1;52 HSD17B1;65
Nuclear non-senile cataract	0.17094268	29	1.447e-03	4.193e-01	ERC6C;412 GJA8;639 GJA3;885 CRYAA;1262 FYCO1;1541 CRYBA2;2128
Renal corticomedullary cysts	0.35012571	7	1.337e-03	4.193e-01	CRB2;630 NHPA;31312 TME6M7;1758 XPNPEP3;2530 NHPA;2629 NHPY1;3518
Renal salt wasting	0.23081853	16	1.393e-03	4.193e-01	CLCNKA;36 CLCNKB;48 CYP11A1;250 KCNJ1;800 STAR;1089 SLC12A1;1555
Cholestasis, chronic	0.40313600	5	1.797e-03	4.254e-01	ATP8B1;113 ABCB4;369 ABCB11;1965 AKR1D1;2062 F2R;3134 NA
Aplasia/Hypoplasia of the cerebellar ver	0.20746270	19	1.747e-03	4.254e-01	CRB1;114 SPECCL1;376 RD3;1244 KCNJ3;1252 RDH12;1499 CEP290;2111
Cerebellar vermis aplasia or hypoplasia	0.20746270	19	1.747e-03	4.254e-01	CRB1;114 SPECCL1;376 RD3;1244 KCNJ3;1252 RDH12;1499 CEP290;2111
Cholangiolocellular Carcinoma	0.37118724	6	1.640e-03	4.254e-01	PROM1;50 ALB;127 CRNK1;12108 CRLF1;2325 MTHFR;2454 LIF;5162
Facial diplegia	-0.28475199	10	1.821e-03	4.254e-01	ADCY6;86 SLC12A2;470 TPM3;1745 SLC12A6;1866 DMPK;2290 TK2;2824